

Question 1

The following steps define an algorithm which operates on two numbers.

- Step 1: Write down the two numbers side by side on the same line.
- Step 2: Beneath the left-hand number write down double the left-hand number.
Beneath the right-hand number write down half of the right-hand number,
ignoring any remainder.
- Step 3: Repeat Step 2 until the right-hand number is 1.
- Step 4: Delete those rows where the number in the right-hand column is even.
Add up the remaining numbers in the left-hand column. This is the result.

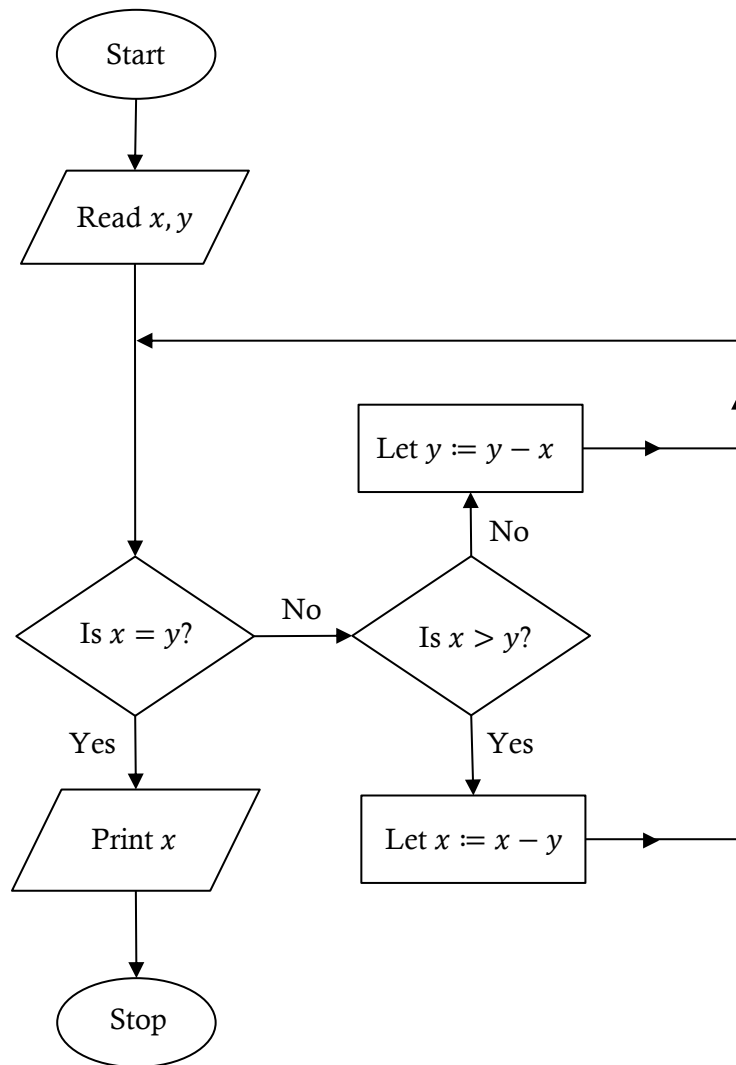
(i) Apply the algorithm to the numbers 50 and 56. [3]

(ii) Use your result from part (i), and any other simpler examples you may choose, to write down what the algorithm achieves. [1]

[Total 4]

ALGORITHMS: FLOWCHARTS AND PSEUDOCODE

1. The following flowchart describes an algorithm which requires two positive integers as inputs.



- a. Apply the algorithm with $x = 45$ and $y = 18$.
b. Apply the algorithm with $x = 84$ and $y = 106$.
c. What does this algorithm achieve? Prove your assertion.
2. The following pseudocode describes the same algorithm. Line numbers are used to aid in navigation. Add the line numbers to the flowchart above to show how the different elements of the algorithm match up.

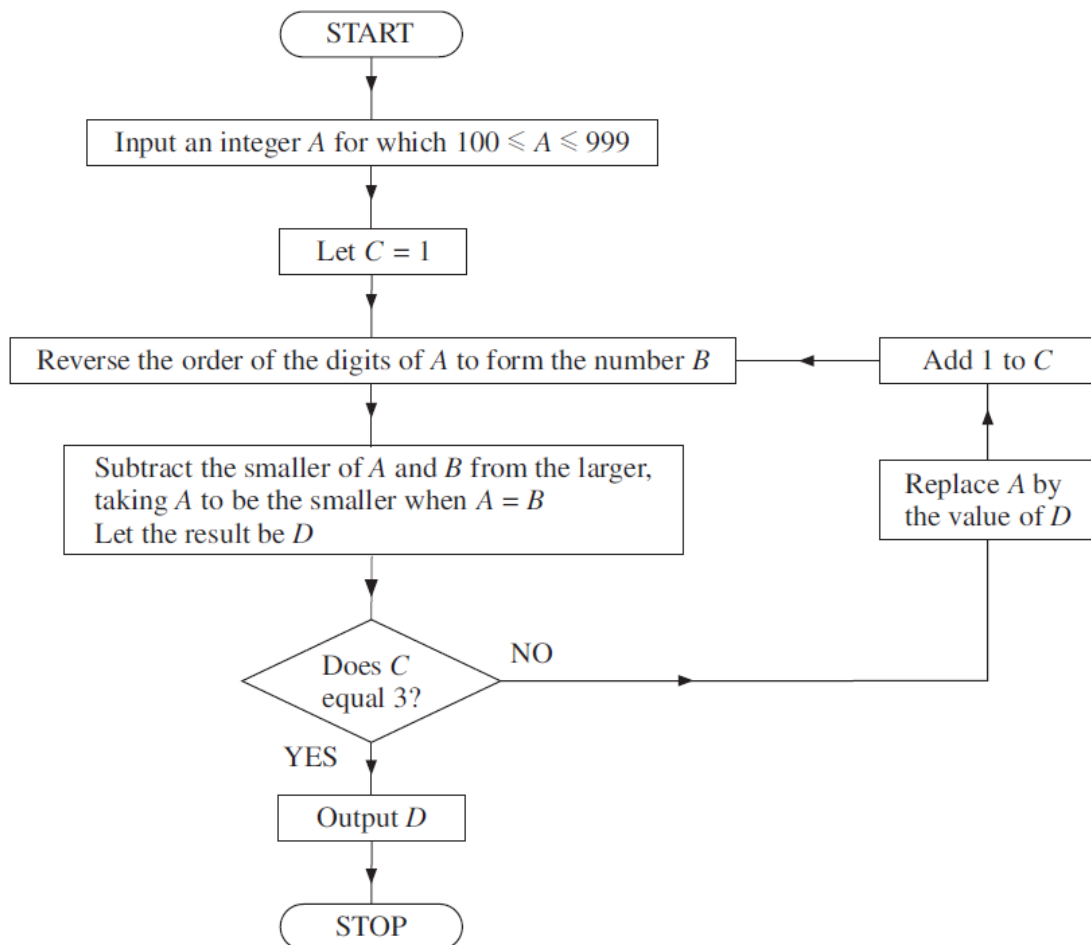
```
10  Read x
20  Read y
30  If  $x = y$  then goto 100
40  If  $x > y$  then goto 70
50  Let  $y := y - x$ 
60  Goto 30
70  Let  $x := x - y$ 
80  Goto 30
100 Print x
110 Stop
```

3. Consider the algorithm below:

STEP 1 Input B
 STEP 2 Input N
 STEP 3 Let $F := N \div B$
 STEP 4 Let $G = \text{INT}(F)$
 STEP 5 Let $H := B \times G$
 STEP 6 Let $C := N - H$
 STEP 7 Output C
 STEP 8 Let $N := G$
 STEP 9 If $N = 0$ then stop
 STEP 10 Go back to STEP 3

- Apply the algorithm with inputs $B = 2$ and $N = 5$. Record the values of F, G, H, C , and N each time STEP 9 is reached.
- Explain what happens when the algorithm is applied with the inputs $B = 2$ and $N = -5$.
- Apply the algorithm with the inputs $B = 10$ and $N = 37$. Record the values of F, G, H, C , and N each time STEP 9 is reached.
- What are the output values when $B = 10$ and N is any positive integer?

4. Consider the following algorithm:



- Trace through the algorithm using the input $A = 614$. Write down the values of A, B, C and D in each pass through the algorithm.
- What is the output when $A = 616$?
- Explain why the counter C is needed.

5. The following algorithm describes the steps of a computer subroutine. It is ‘called’ using the command $F(n)$ where n is the input of the subroutine, which must be a positive integer. This algorithm is recursive, because it calls upon itself.

STEP 1 If $n = 1$ then let $m := 1$.

STEP 2 If $n \neq 1$ then let $m := n \times F(n - 1)$

STEP 3 Output m .

What does this recursive algorithm achieve?

6. The following steps define an algorithm which acts on two positive integers.

STEP 1 Write down the two numbers side by side on the same row.

STEP 2 Beneath the left-hand number write down double that number. Beneath the right-hand number write down half of that number, ignoring any remainder.

STEP 3 Repeat STEP 2 until the right-hand number is 1.

STEP 4 Delete those rows where the right-hand number is even. Add up the remaining left-hand numbers. This is the result.

- a. Apply the algorithm with left-hand number 3 and right-hand number 8.
 - b. Apply the algorithm with left-hand number 26 and right-hand number 42.
 - c. Use these results, together with any other examples you may choose, to write down what the algorithm achieves. Prove your assertion.
 - d. Write the algorithm more formally using instructions that could be read by a computer. Use variables for the two numbers, and write the instructions in STEP 2 using formulae. You will need to find a way to implement STEP 3 and STEP 4 using conditional statements and variables.
7. Write your own algorithm to convert a positive integer into binary. Use a flowchart to describe your algorithm.
8. Write an algorithm to test whether a positive integer is prime. Describe your algorithm using pseudocode.
9. Produce a flow chart to show how the next-fit algorithm works.